2. Implement a basic finite state automaton that recognizes a specific language or pattern. In this example, we'll create a simple automaton to match strings ending with 'ab' using python.

Aim: To implement a finite state automaton that accepts strings ending with ab.

Procedure:

1. Read an input string containing a and b.
2. Initialize the state.
3. Update the state for each character.
4. Check the final state.
5. Display whether the string is accepted or rejected.

Code:

```
def check_string(s):
    state = 0
    for ch in s:
        if state == 0:
            if ch == 'a':
                state = 1
        elif state == 1:
            if ch == 'a':
                state = 1
            elif ch == 'b':
                state = 2
        elif state == 2:
            if ch == 'a':
                state = 1
            elif ch == 'b':
                state = 0
```

```
    if state == 2:
        print("Accepted")
    else:
        print("Rejected")
check_string("abab")
check_string("aaab")
check_string("abba")
```

Output:

```
C:\Users\blohi\Downloads\NLP lab>C:\Users\blohi\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/b
lohi/Downloads/NLP lab/2.py"
Accepted
Accepted
Rejected
```

Result: The finite state automaton successfully recognizes strings ending with ab.